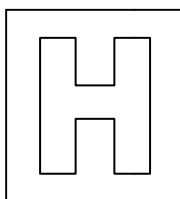


Candidate Name: _____

Class Adm No

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2014 Preliminary Examination II

Pre-university 3

H2 Biology

9648/02

Paper 2

16 September 2014

2 hours

Additional Materials: Writing paper

READ THESE INSTRUCTIONS FIRST

Do not open this booklet until you are told to do so.

Write your Index number and name on all the work you hand in.
Write in dark blue or black pen on both sides of the paper.
You may use a soft pencil for any diagrams, graphs or rough working.
Do not use staples, paper clips, highlighters, glue or correction fluid.

Section A

Answer **all** questions on the question booklet.

Section B

Answer **one** question on writing paper.

At the end of the examination, fasten all your work securely together.
The number of marks is given in brackets [] at the end of each question or part question. At the end of the examination, fasten all your work securely together.

For Examiner's Use	
Section A	
1	
2	
3	
4	
5	
6	
Section B	
Total	

This question paper consists of 18 printed pages.

[Turn over

Section A

Answer **all** questions in this section.

- 1** The fluid mosaic model of membrane structure was proposed in 1972 to describe in detail how the components of the cell membrane are organised.

(a) Some components of the cell membrane include:

- phospholipid molecules
- extrinsic protein molecules
- intrinsic protein molecules
- cholesterol molecules

- (i)** In the space below, draw a labelled diagram of a cross-section of the cell surface membrane to show how the components above are organised.

Label the inner and outer layers of the membrane.

[5]

- (ii)** Suggest why 'fluid mosaic' is an appropriate term to use to describe the structure of the cell surface membrane.

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.....
..... [2]

The insulin receptor is embedded in the cell surface membrane of specific cell types. Insulin, secreted by the alpha cells of the Islets of Langerhans, is circulated through the blood, binding to target cells that have insulin receptors.

The oestrogen receptor is found in the cytoplasm of cells.

- (b)** Explain why the insulin receptor is found on the cell surface whereas the oestrogen receptor is found in the cell cytoplasm.

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..... [2]

- (c)** Explain how binding of insulin to its receptor results in lowering of blood glucose levels.

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..... [4]

- (d)** Suggest how amplification may occur in the insulin signalling pathway.

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..... [1]

[Total: 14]

- 2 Fig. 2.1 shows the assembly of the transcription initiation complex at the promoter of a proto-oncogene.

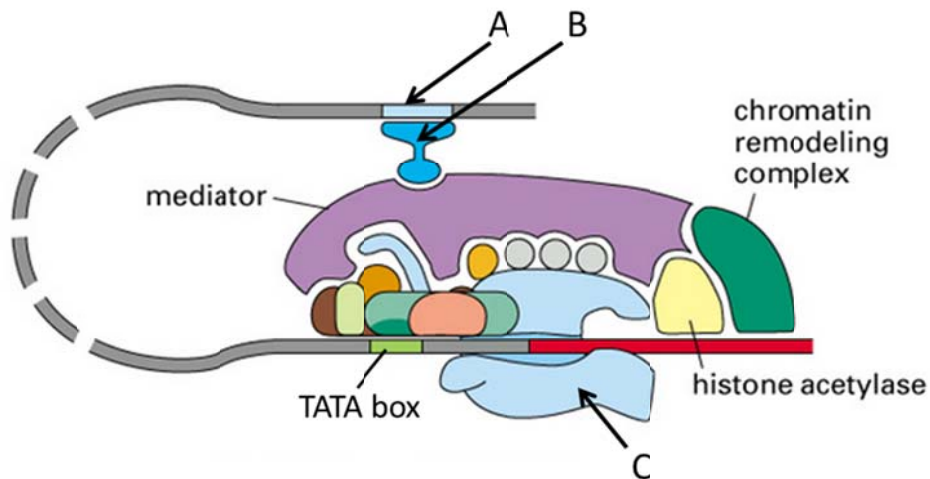


Fig. 2.1

- (a) Identify the components labelled **A**, **B** and **C**.

A:

B:

C: [1]

A mutation in the gene coding for one of the components shown in Fig. 2.1 results in the conversion of the proto-oncogene into an oncogene.

- (b) Suggest which component in Fig. 2.1 the mutation could have affected and explain how the mutation could result in cancer.

.....

 [4]

- (c) Suggest two differences between transcription and translation.

.....

 [2]

- (d) DNA polymerase is an enzyme involved in the replication of DNA. Ara-ATP is a chemical that is able to affect the enzymatic activity of DNA polymerase. Fig. 2.2 shows the structure of ara-ATP.

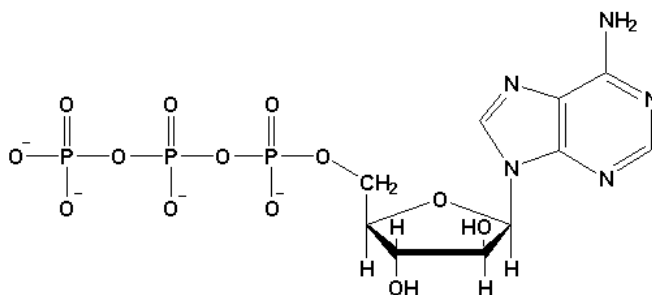


Fig. 2.2

A student decided to investigate the effect of different concentrations of ATP on the rate of DNA synthesis. The results of his experiment are shown in Fig. 2.3.

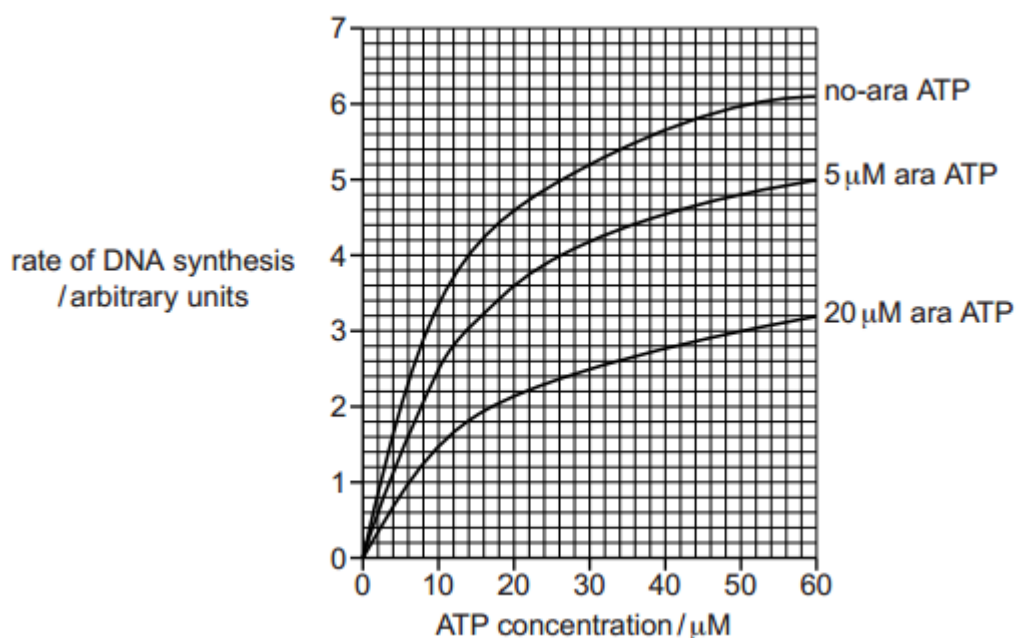


Fig. 2.3

- (i) With reference to Fig. 2.3, describe the effect of ara-ATP on DNA polymerase.

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..... [2]

- (ii) Explain how ara-ATP affects DNA polymerase activity, in terms of the mode of action of enzymes.

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..... [3]

- (iii) State the names of two other enzymes required for DNA replication and explain their roles.

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..... [4]

- (e) Explain why there is a leading and lagging strand during DNA replication.

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..... [2]

[Total: 18]

- 3** Woolly mammoths are extinct mammals that are related to elephants of the *Mastodon* genus. Woolly mammoths existed around 200,000 years ago during the Ice Age, where the climate changed and temperatures in the Arctic drastically decreased. These mammoths had various adaptations to reduce heat loss, such as having thick fur, small ears and small tails. During the Ice Age, the population of the woolly mammoths was significantly larger than the population of mastodons.

Woolly mammoths went extinct approximately 4000 years ago, as a result of climate change. However, the population of mastodons increased in number.

- (a)** Explain how natural selection resulted in the population changes of the woolly mammoths and the mastodons.

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..... [3]

Scientists have used various methods to investigate the relatedness between Asian elephants and woolly mammoths.

In the first investigation, they compared the structural features of these two species, finding that both animals have similar bone structure and tusks. In the second investigation, they compared the amino acid sequence of haemoglobin from both animals.

- (b)** Explain which of the two investigations is likely to provide a more accurate determination of the relatedness of these two species.

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..... [2]

Scientists studied the haemoglobin of woolly mammoths and Asian elephants to determine whether or not the differences in haemoglobin were part of the mammoth's adaptation to a cold climate.

It is advantageous for Arctic mammals to have haemoglobin whose affinity for oxygen is not significantly affected by changes in temperature. This is achieved by using substances called 'red cell effectors', which can bind to haemoglobin.

Fig. 3.1 shows the effect of temperature on the affinity of oxygen of woolly mammoth and Asian elephant haemoglobin, with and without red cell effectors.

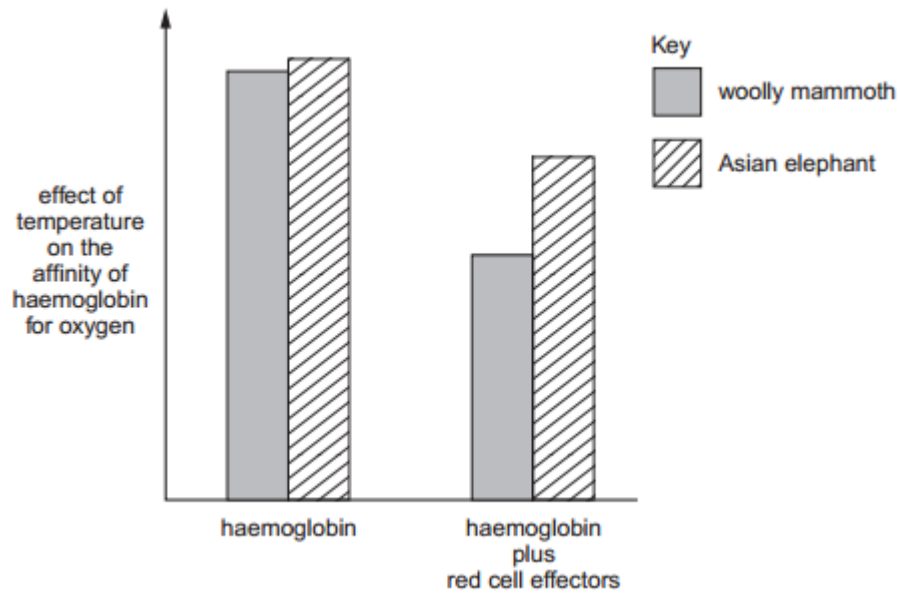


Fig. 3.1

- (c) With reference to Fig. 3.1, explain whether or not there is evidence that woolly mammoths are better adapted for a cold climate than Asian elephants.

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..... [3]

[Total: 8]

- 4** In Bengal tigers, coat colour is controlled by one gene while eye colour is controlled by a separate gene. When an orange tiger with brown eyes mates with a white tiger with blue eyes, all the offspring have orange coat colour and blue eyes.
- (a)** Draw a genetic diagram to show the result of a cross between two tigers of the F_1 generation.

[4]

- (b) Explain how it would be possible to determine whether or not an offspring from the cross in (a) with orange coat and blue eyes is heterozygous or homozygous dominant for both traits.

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..... [3]

- (c) A wild-life researcher conducted an investigation into the numbers of offspring for the cross conducted in (a). The numbers of offspring in the F₂ generation he observed are as follows:

Orange coat with blue eyes	16
Orange coat with brown eyes	2
White coat with blue eyes	4
White coat with brown eyes	10

$$\chi^2 = \sum \frac{(O-E)^2}{E}$$

Table 4.1

degrees of freedom	probability, <i>p</i>				
	0.10	0.05	0.02	0.01	0.001
1	2.71	3.84	5.41	6.64	10.83
2	4.61	5.99	7.82	9.21	13.82
3	6.25	7.82	9.84	11.35	16.27
4	7.78	9.49	11.67	13.28	18.47

- (i) Using the formula and Table 4.1 provided above, conduct a chi-squared test to determine whether or not the observed numbers in the F₂ generation differ significantly from the expected ratio.

- (ii) Suggest an explanation for the results you have obtained in part (i).

.....
[1]

The tiger, *Panthera tigris*, is classified as an endangered species by the International Union for the Conservation of Nature and Natural Resources (IUCN).

- (d) Complete Table 4.2 to show the classification of the tiger.

Table 4.2

Kingdom	
	Chordata
Class	Vertebrata
	Mammalia
	Carnivora
Genus	
	<i>Panthera tigris</i>

[3]

- (e) Fig. 4.1 shows the number of tigers in the wild between 1900 and 2010.

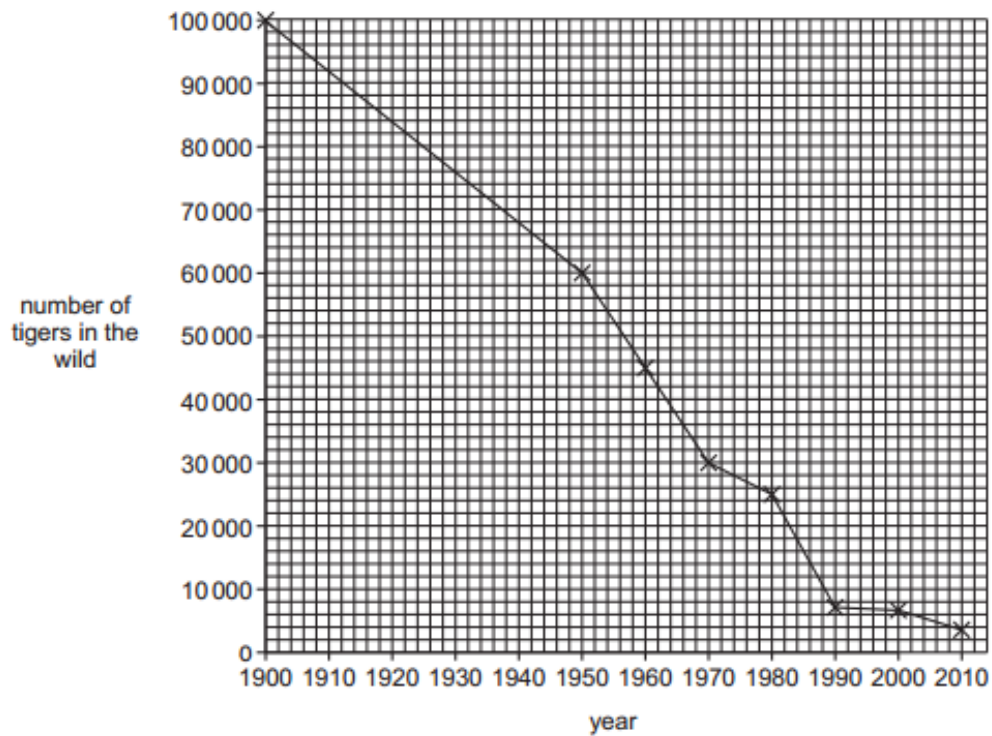


Fig. 4.1

- (i) Calculate the steepest rate of decrease observed in the tiger population between 1900 and 2010.

..... tigers per year [1]

- (ii) Suggest a reason for the steep decrease that you have identified in part (i).

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..... [1]

[Total: 16]

- 5 Table 5.1 shows the mean axon diameter and mean speed of conduction of nerve impulses for four different animals.

Table 5.1

animal		myelination of neuron	axon diameter / μm	mean speed of conduction / ms^{-1}
mammal	rabbit	myelinated	3	30
	bear	non-myelinated	4	5
amphibian	turtle	myelinated	20	121
	frog	myelinated	12	40

- (a) With reference to Table 5.1, describe and explain:

- (i) the effect of axon diameter on the speed of conduction of impulses in amphibians.

.....

 [3]

- (ii) the effect of myelination on the speed of conduction of impulses in mammals.

.....

 [3]

- (b)** Excessive breeding of pigeons may become an environmental issue as pigeons are able to spread disease and lower hygiene. In order to reduce the pigeon population, the practice of culling pigeons is carried out by environmental agencies, often through the use of chemicals such as Avitrol.

Avitrol is the commercial name for the chemical 4-aminopyridine. This chemical functions as an inhibitor of voltage-gated K^+ channels. Upon exposure of pigeons to Avitrol, the pigeon experiences epileptic-like convulsions.

- (i)** Explain how Avitrol will affect transmission of an action potential.

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..... [3]

Besides inhibiting the voltage-gated K^+ channels, Avitrol can also increase the release of acetylcholine at cholinergic synapses.

- (ii)** Explain how the effect of Avitrol at synapses will lead to muscle convulsions.

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..... [3]

[Total: 12]

- 6 Fig. 6.1 shows an electron micrograph of several plant cells.

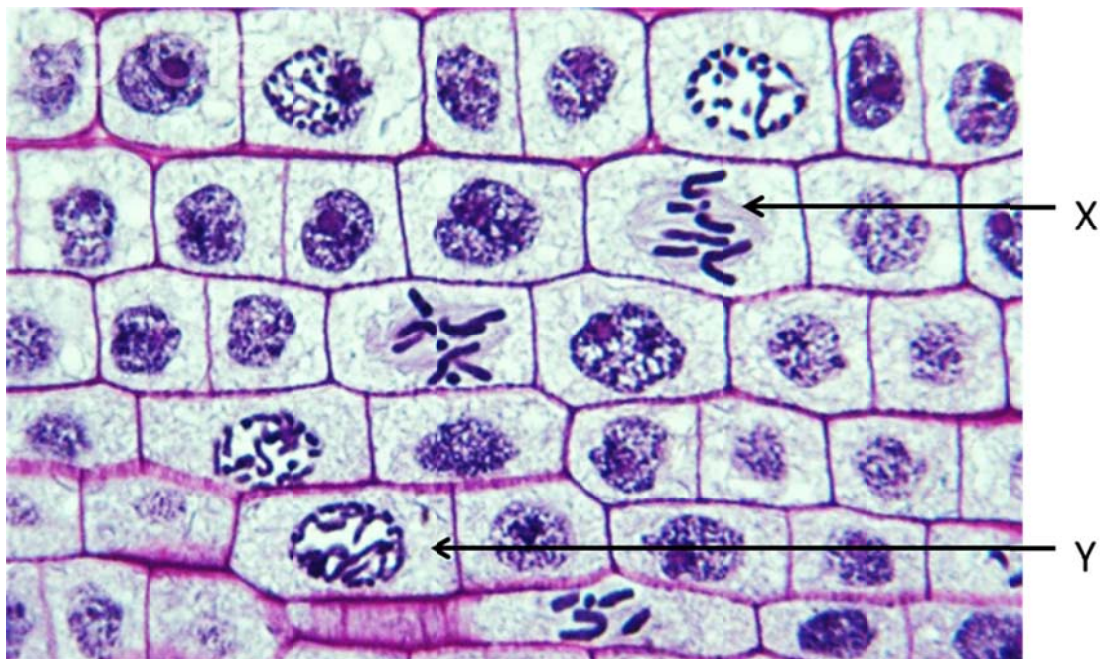


Fig. 6.1

- (a) Identify the stages of mitosis shown in cells X and Y.

X:

Y: [1]

- (b) Describe the events that occur during the stages of the cell cycle identified in cells X and Y.

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..... [4]

- (c)** Explain the process of cytokinesis in plants.

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..... [3]

- (d)** Colchicine is a drug that is frequently used on crops to induce formation of polyploid plants.

- (i)** State the meaning of the term 'polyploid'.

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..... [1]

- (ii)** Explain how colchicine may result in the formation of polyploid plants.

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..... [3]

[Total: 12]

Section B

Answer **one** question.

Write your answers on the separate answer paper provided.
Your answers should be illustrated by large, clearly labelled diagrams, where appropriate.
Your answers must be in continuous prose, where appropriate.
Your answers must be set out in sections **(a)**, **(b)** etc., as indicated in the question.

EITHER

- 7 **(a)** Explain how the structure of the chloroplast contributes towards its function in carbohydrate synthesis. [6]
- (b)** Compare and contrast between the processes of oxidative phosphorylation and photophosphorylation. [6]
- (c)** Discuss how structure is related to function for starch and cellulose. [8]

[Total: 20]

OR

- 8 **(a)** Explain how the reproductive cycle of the bacteriophage allows it to contribute towards genetic variation in bacteria. [6]
- (b)** Explain how the presence of lactose and glucose can affect bacterial growth. [10]
- (c)** Distinguish between the reproductive cycles of the T4 phage and influenza virus. [4]

[Total: 20]

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